

HelixCode Feature Status

Revision	4
Created	2026-06-15
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Status	active (population in progress)
Status summary	docs/features/Status_Summary.md
Continuation	docs/CONTINUATION.md

Authoritative, in-depth inventory of **every** HelixCode feature across all services, infrastructure, and client applications — including capabilities ported from the `cli_agents/` reference catalogue — with per-feature status across every dimension the operator mandate (2026-06-15) requires. Kept in sync via the `docs_chain` engine (§11.4.106, `submodules/docs_chain`) and the Status-doc covenant (§11.4.45 / §11.4.53 / §11.4.56 / §11.4.57 / CONST-063 / CONST-064).

Anti-bluff (CONST-035 / §11.4.83 / §11.4.107): a feature is marked video-confirmed (📺 yes) ONLY when a real recorded scenario in `/Volumes/T7/Downloads/Recordings` shows it working end-to-end with a strong real LLM and that recording has been analyzed. No false "yes". An un-recorded or un-analyzed feature is honestly 📺 no / 📺 pending, never bluffed green.

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Status dimensions (legend)

Each feature row carries:

Column	Meaning	Values
Area	service / infrastructure / application(client) / submodule	—
Component	package / tool / app / submodule it lives in	—
Feature	the discrete user-or-system capability	—
Dev	implementation status	

Column	Meaning	Values
		done / partial / stub / absent
Wired	reachable from a shipped flow (not dead code)	yes / no / partial
Real-use	genuinely usable by an end user	yes / no / unknown
Tests	automated coverage	unit / integ / e2e / none (combinable)
V&V	captured runtime evidence (§11.4.5/§11.4.69)	yes(path) / no
 Video	recorded real scenario + analyzed (§11.4.83/§11.4.107)	yes(path) / pending / no / n/a
Analysis	comprehensive recording analysis performed	yes / no
Origin	native / ported:<cli_agent>	—
Overall	rollup	confirmed / working-untaped / partial / gap

Population progress

This document is populated by background inventory subagents fanning out across the codebase (§11.4.70 subagent-driven, §11.4.103 parallel-streams). Coverage is reported honestly — confirmed rows require a real analyzed video; everything else is marked truthfully. Population is an ongoing program, NOT a one-shot claim.

Slice	Scope	Status
internal services + infra	helix_code/internal/* (72 pkgs)	inventory dispatched
cmd tools + client apps	helix_code/cmd/* (21) + applications/* (cli/tui/web/desktop/mobile)	inventory dispatched
owned submodules	submodules/* (70+)	inventory dispatched
ported cli_agents capabilities	cli_agents/* → HelixCode	inventory dispatched

Feature inventory

Aggregated from docs/features/inventory/.md by scripts/generate_features_status.sh (docs_chain §11.4.106).*

CLI / TUI / Web / Desktop / Mobile clients + cmd tools

Inventory slice for `helix_code/cmd/*` (cmd tools) and `helix_code/applications/*` + `helix_code/web/*` (client applications). Assessed from source evidence per the anti-bluff covenant (CONST-035 / §11.4.107). 📹 **Video is no for every row** — device/display recordings are only just starting; the conductor owns video confirmation. Overall is therefore never confirmed; honest rollups are working-untaped / partial / gap.

Recordability per client (for the conductor): **CLI, TUI, Web = feasible now** (terminal + headless HTTP). **Desktop (Fyne) = host-display required**. **Mobile:** Aurora OS / Harmony OS are Go/Fyne desktop-class binaries (recordable on a Linux display); **Android / `aurora_os` HAP / `harmony_os` HAP need device/emulator; iOS needs a built app (no Xcode project present → not buildable yet).**

Area	Component	Feature	Dev	Wired
application(cli)	cmd/cli REPL	Plain-text prompt → real LLM (streaming via <code>provider.GenerateStream</code>)	done	yes
application(cli)	cmd/cli REPL	<code>/generate</code> real LLM generation (BLUFF-001 resolved)	done	yes
application(cli)	cmd/cli REPL	<code>/models</code> list models via <code>providerManager.GetProviders</code> (BLUFF-002 resolved)	done	yes
application(cli)	cmd/cli REPL	<code>/workers</code> list workers	done	yes
application(cli)	cmd/cli REPL	<code>/health</code> health check	done	yes
application(cli)	cmd/cli REPL	<code>/diff [ref]</code> real git diff (os/exec)	done	yes
application(cli)	cmd/cli REPL	<code>/debate <prompt></code> DebateOrchestrator over real provider	done	yes
application(cli)	cmd/cli REPL	<code>/specify <request></code> HelixSpecifier Specify phase over real provider	done	yes
application(cli)	cmd/cli REPL	<code>/checkpoint [create/list/restore]</code> workspace snapshots	done	yes
application(cli)	cmd/cli REPL	<code>/undo</code> revert last action	done	yes

Area	Component	Feature	Dev	Wired
application(cli)	cmd/cli REPL	/clear /reset clear conversation history	done	yes
application(cli)	cmd/cli REPL	/help help; /exit /quit exit	done	yes
application(cli)	cmd/cli REPL	@-file mentions (real os.ReadFile context attach)	done	yes
application(cli)	cmd/cli REPL	Context-window tracking + generation stats	done	yes
application(cli)	cmd/cli commands	list / show / run / reload markdown commands	done	yes
application(cli)	cmd/cli hooks	list / enable / disable / validate / test git hooks	done	yes
application(cli)	cmd/cli lsp	status / list-servers / restart / stop LSP servers	done	yes
application(cli)	cmd/cli mcp	add / remove / list / test / auth / logs MCP servers	done	yes
application(cli)	cmd/cli permissions	list / add / remove / check tool permissions	done	yes
application(cli)	cmd/cli sessions	list / show / delete sessions	done	yes
application(cli)	cmd/cli skills	list / show / invoke / reload skills	done	yes
service	cmd/server	HTTP/gRPC/WebSocket server boot (bin/helixcode)	done	yes
application(web)	web/frontend	LLM generate console (POST /api/v1/llm/generate)	done	yes
application(web)	web/frontend	LLM streaming console (SSE, POST /api/v1/llm/stream)	done	yes
application(web)	web/frontend	Specify phase form (POST /api/v1/specify)	done	yes
application(web)	web/frontend	Response/metadata rendering (no client simulation)	done	yes
service	internal/server (HTTP API)	/api/v1/llm/generate real provider.Generate	done	yes

Area	Component	Feature	Dev	Wired
service	internal/server (HTTP API)	/api/v1/llm/stream real provider.GenerateStream (SSE)	done	yes
service	internal/server (HTTP API)	/api/v1/specify real speckit Specify phase	done	yes
service	internal/server (HTTP API)	/api/v1/llm/providers, /llm/ models list	done	yes
service	internal/server (HTTP API)	/api/v1/auth register / login / logout / refresh (JWT)	done	yes
service	internal/server (HTTP API)	/api/v1/users me get/update/delete	done	yes
service	internal/server (HTTP API)	/api/v1/workers CRUD + heartbeat + metrics	done	yes
service	internal/server (HTTP API)	/api/v1/tasks CRUD + assign/start/complete/fail/retry/checkpoint	done	yes
service	internal/server (HTTP API)	/api/v1/projects CRUD + sessions + planning/building/testing/refactoring workflows	done	yes
service	internal/server (HTTP API)	/api/v1/sessions CRUD	done	yes
service	internal/server (HTTP API)	/api/v1/system stats + status	done	yes
service	internal/server (HTTP API)	/api/v1/memory systems + stats	done	yes
service	internal/server (HTTP API)	/api/v1/qa session start/list/status/report/screenshot/cancel	done	yes
service	internal/server (HTTP API)	/api/v1/screenshot engines + capture	done	yes
service	internal/server (HTTP API)	/ws MCP WebSocket bridge	done	yes
service	internal/server (HTTP API)	/health, /api/v1/health, /metrics, /api/v1/server/info	done	yes
service	internal/server (HTTP API)	/debug/pprof/* (opt-in profiling)	done	yes
application(tui)	applications/terminal_ui	LLM chat (real provider, verifier-driven model discovery)	done	yes

Area	Component	Feature	Dev	Wired
application(tui)	applications/terminal_ui	Dashboard / Tasks / Workers / Projects / Sessions panels (real managers)	done	yes
application(tui)	applications/terminal_ui	Sidebar nav + key bindings (d/t/w/p/s/l/q/c)	done	yes
application(tui)	applications/terminal_ui	Skill dispatcher + tool registry (git/fs/grep/LSP/MCP, graceful-nil)	done	yes
application(tui)	applications/terminal_ui	HelixMemory durable cross-session store (SQLite, default-on)	done	yes
application(tui)	applications/terminal_ui	Status bar (DB/Redis/LLM status, context %), notifications, themes, i18n	done	yes
application(tui)	applications/terminal_ui	QA engine panel (helixqa.Engine wired)	partial	yes
application(desktop)	applications/desktop (Fyne)	LLM chat tab (real provider, verifier-driven models)	done	yes
application(desktop)	applications/desktop (Fyne)	Dashboard / Tasks / Workers / Projects / Sessions tabs (real managers)	done	yes
application(desktop)	applications/desktop (Fyne)	Settings tab (theme, server config, shortcuts; desktop.yaml)	done	yes
application(desktop)	applications/desktop (Fyne)	Agentic tools + skills/plugins wiring (graceful-nil)	done	yes
application(desktop)	applications/desktop (Fyne)	NoGUI build mode (-tags nogui CLI fallback)	done	yes
application(desktop)	applications/desktop (Fyne)	TUI-parity wiring (parity_wiring_test.go)	done	yes
application(mobile)	applications/aurora_os (Go/Fyne)	Full multi-page GUI: dashboard/projects/sessions/tasks/LLM/workers/system	done	yes
application(mobile)	applications/aurora_os (Go/Fyne)	NoGUI CLI mode (cobra) + theme system	done	yes
application(mobile)	applications/harmony_os (Go/Fyne)	10-tab GUI incl. distributed engine + multi-device scheduling	done	yes

Area	Component	Feature	Dev	Wired
application(mobile)	applications/harmony_os (Go/Fyne)	NoGUI CLI + interactive shell + theme system	done	yes
application(mobile)	applications/android (Kotlin)	Connect + task list (RecyclerView) over Go mobile-core bridge	partial	partial
application(mobile)	applications/android (Kotlin)	Models / settings / notifications / theme UI	stub	no
application(mobile)	applications/ios (Swift)	Connect + task list (UITableView) over Go mobile-core bridge	partial	partial
application(mobile)	applications/ios (Swift)	Models / settings / notifications / theme UI	stub	no
application(cli)	cmd/helix_config	Interactive provider/credential config wizard → YAML/.env	done	yes

Area	Component	Feature	Dev	Wired
application(cli)	cmd/config_test	Config + provider-credential validator	done	yes
application(cli)	cmd/i18n	i18n bundle/translator tooling (library, no main)	done	yes
application(cli)	cmd/infrastructure	Container / k8s / registry readiness checks	done	yes
application(cli)	cmd/performance_optimization	pprof profiling + bottleneck analysis + benchmarks	done	yes
application(cli)	cmd/security_test	Security test harness (PoC execution + verification)	done	yes
application(cli)	cmd/security_fix	Finding ingestion + policy-driven fix + re-scan validation	done	yes
application(cli)	cmd/security_fix_standalone	Batch parallel security fix + audit-trail reporting	done	yes
application(cli)	cmd/security_scan	AST/pattern code scan + leak detect + SARIF report	done	yes

Honesty notes (anti-bluff)

- **cmd/security_scan has ZERO tests** (only main.go) — marked Tests=none, Overall=gap. Untested scanner = bluff risk per CONST-048/CONST-050.
- **iOS / Android are scaffolds**: real source (JSON parse, list bind, Go mobile-core bridge) but single-screen, hardcoded localhost test server, **no Xcode project / no Gradle+manifest** → not buildable, not recordable. Real=use=no, Overall=gap.
- **Aurora OS / Harmony OS are genuine Go/Fyne apps** (buildable via make aurora-os / make harmony-os, comprehensive unit+integration tests) — but integration features need real PostgreSQL/Redis/LLM backends, and full HAP/ multi-device exercise needs the actual OS environment.
- **Web LLM endpoints** (/llm/generate, /llm/stream, /specify) have real Ollama-backed e2e tests (tests/integration/{llm_generate, llm_stream, specify_server}_e2e_test.go, build tag integration, honest SKIP-OK when Ollama unreachable) → Tests=e2e, V&V=yes (docs/qa/web-llm-e2e-20260615/). The 60+ CRUD/auth/workflow endpoints are real but tested at the manager/service layer, not at the HTTP-transport layer → Tests=integ/none honestly.
- **No row is confirmed** — every 📺 Video=no; rollups are working-untaped (real + tested, no video), partial (real but thin/unverified test coverage), or gap (scaffold or untested).
- **Themed brand re-records (Rev 3, 2026-06-15)**: the logo+dark-theme (lime #A8DD22 + teal #8FC9B8, from assets/Logo.png) was applied to all 6 clients (commit f9c6ab06) and **5 of 6 clients now have a NEW themed, real-LLM/real-data, window-scoped re-recording** referenced in their Analysis cells: helixcode-cli-themed-20260615.mp4 (CLI: brand banner + real DeepSeek "2+2 equals 4"; 790x560, 4.9s), helixcode-web-themed-20260615.mp4 (Web: dark theme + logo + real DeepSeek, viewport-scoped 1280x800, 5.2s), helixcode-tui-themed-20260615.mp4 (TUI: brand tview.Styles + banner + real DeepSeek streamed answer; terminal-pane-scoped, 29.8s), helixcode-desktop-themed-20260615.mp4 (Desktop: themed dashboard + embedded

logo, window-scoped via screencapture -l, 144 frames, liveness via advancing uptime — GUI-interaction-driving is operator-attended §11.4.52 as Fyne/OpenGL rejects synthetic input), helixcode-android-themed-20260615.mp4 (Android: Material dark + logo + real connect + real server task list; device-scoped 1080x2424, 10.4s). All window-scoped per §11.4.154 (terminal pane / browser viewport / app window / device frame — never whole-desktop), with §11.4.154 fresh-corpus rotation honored. **iOS themed re-record is OPERATOR-BLOCKED** (§11.4.52, honest SKIP not bluff): the themed HelixTheme.swift (lime/teal) + logo asset are present, Swift-compile cleanly, and the real Go-core wiring (CoreNewMobileCore/createTask) is genuine — but a window-scoped simulator video cannot be captured because CoreSimulatorService (the sandboxed daemon) is denied write access to the external /Volumes/T7 volume for asset-catalog/IB device-set creation (FACT: NSCocoaErrorDomain 513 in ~/Library/Logs/CoreSimulator/CoreSimulator.log). Unblock (host TCC change, out of agent scope): grant CoreSimulatorService Full-Disk-Access to /Volumes/T7, OR relocate Xcode's IB-Support/DerivedData device-set to a local APFS path. No fabricated video was produced; the prior helixcode-ios-launch-20260615.mp4 was NOT rotated (no replacement). 5 of 6 clients are real-LLM themed-recorded; the 6th awaits this host unblock.

Feature count: 67 rows (CLI REPL 14, CLI subcommand groups 7, server boot 1, web frontend 4, HTTP API groups 18, TUI 7, desktop 6, mobile 8, other cmd tools 11 + security_scan 1).

Deepened inventory (round 2)

Round-2 deepening of packages flagged shallow/partial/unknown in round 1 (internal_services.md coverage notes + the owned-submodule umbrella rows). Each previously-thin entry is enumerated into discrete features in the SAME 12-column schema as internal_services.md (legend in docs/features/Status.md § "Status dimensions"). Assessed from real code evidence (impl reality, wiring, *_test.go presence) per CONST-035 / §11.4.107 anti-bluff.

Anti-bluff caveats (read before trusting a row).

- 📺 Video is **no for every row** — no analyzed recording exists; the conductor owns video confirmation. Overall is therefore **never confirmed**; honest rollups are working-untaped / partial / gap.
- Origin is native (all own-org code).
- **Round-1 stub claims corrected here from direct source reads** (a round-2 exploration subagent reported these as stubs WITHOUT reading the code — verified false): workspace docker Stop/Remove are real (docker stop / docker rm -f, manager.go:21-37); voice recorder Stop is real (Process.Signal(os.Interrupt) then Kill(), recorder.go:63-80); roocode CodeReviewer.Review is real (reads the file + scans for TODO/issue patterns, reviewer.go:16-30). They are done/partial (impl real) but Real-use=unknown (shipped-flow wiring unconfirmed), not stubs.
- Submodule-side Wired follows internal_services.md's wiring model: helix_agent is imported directly (Wired=yes where HelixCode consumes it), transitively-reachable own-org deps are Wired=partial, repo-present-but-not- in-build-graph are Wired=no. helix_specifier/

- helix_qa are imported by module path (Wired=yes); panoptic is a Challenge/recording submodule not in HelixCode's Go build graph (Wired=no).
- Submodule Tests reflects *_test.go presence in/near the package; it is NOT a coverage-percentage claim. README-cited coverage figures (helix_agent "65.6%", panoptic "78%") are NOT used as evidence — unverified.

Internal packages (round-1 partial/unknown → deepened)

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
infrastructure	internal/clientcore	agentic tool registry wiring (git/fs/LSP/MCP, live-built)	done	yes	yes	unit	no
infrastructure	internal/clientcore	LSP read-only diagnostics tool wiring	done	yes	unknown	unit	no
infrastructure	internal/clientcore	MCP config-merge + server startup + tool registration	done	yes	unknown	unit	no
infrastructure	internal/clientcore	tool-loop system-prompt generation (from registry names)	done	yes	yes	unit	no
infrastructure	internal/clientcore	tool-trace adapter (agent → ensembleui)	done	yes	yes	unit	no
infrastructure	internal/clientcore	skills + plugins loader (graceful-on-failure)	done	yes	unknown	unit	no
infrastructure	internal/clientcore	verifier adapter wiring (CONST-036/037 single-source)	done	yes	unknown	unit	no
infrastructure	internal/clientcore	env-provider registration (cloud API keys, zero-hardcode)	done	yes	unknown	unit	no
infrastructure	internal/clientcore	HelixAgent provider registration (when server reachable)	done	partial	unknown	unit	no
infrastructure	internal/agentbridge	verifier bridge config wrapper	done	yes	unknown	unit	no
infrastructure	internal/agentbridge	VerifyModel real HTTP call to verifier service	done	yes	unknown	unit	no
infrastructure	internal/agentbridge	test request/result marshal+unmarshal	done	yes	unknown	unit	no
infrastructure	internal/checkpoint	git-backed snapshot (stash/write-tree/read-tree)	done	yes	yes	unit	no
infrastructure	internal/checkpoint	file-copy fallback backend (real os.Read/Write)	done	yes	yes	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
infrastructure	internal/ checkpoint	checkpoint create (capture)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	checkpoint restore (undo, writes real bytes)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	metadata persistence (label/timestamp JSON)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	multi-backend auto-selection (git else files)	done	yes	yes	unit	no
service	internal/ ensembleui	tool-trace formatting (text/markdown render)	done	yes	unknown	unit	no
service	internal/ ensembleui	metadata extraction helpers (int/string/slice)	done	yes	unknown	unit	no
service	internal/ ensembleui	multi-model response assembly	done	yes	unknown	unit	no
service	internal/ ensembleui	per-model token-usage aggregation	done	yes	unknown	unit	no
infrastructure	internal/ substrate	Unit interface (task contract)	done	yes	unknown	unit	no
infrastructure	internal/ substrate	queue-based scheduler (enqueue + dispatch)	done	yes	unknown	unit	no
infrastructure	internal/ substrate	serial FIFO execution (single worker)	done	yes	unknown	unit	no
infrastructure	internal/ substrate	context-cancellation propagation to Execute	done	yes	unknown	unit	no
infrastructure	internal/ workspace	workspace create (project bootstrap)	done	partial	unknown	unit	no
infrastructure	internal/ workspace	docker runner start/stop/remove/list (real os/exec)	done	partial	unknown	unit	no
infrastructure	internal/ workspace	workspace status tracking (enum + DB persist)	done	partial	unknown	unit	no
infrastructure	internal/ workspace	CLI tools (create/delete/list registration)	done	yes	unknown	unit	no
infrastructure	internal/ workspace	i18n translator seam	done	yes	yes	unit	no
infrastructure	internal/ workspace	auto-cleanup / TTL teardown	partial	no	unknown	unit	no
service	internal/ voice	audio device detection (arecord/sox/parec probe)	done	partial	unknown	unit	no
service	internal/ voice	audio capture start (real exec.Command)	done	partial	unknown	unit	no
service	internal/ voice	audio capture stop (Signal(Interrupt)→Kill)	done	partial	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
service	internal/voice	Whisper API transcription (real multipart HTTP POST)	done	partial	unknown	unit	no
service	internal/voice	WAV validation before transcribe	done	partial	unknown	unit	no
service	internal/voice	local whisper.cpp fallback transcription	partial	no	unknown	unit	no
service	internal/roocode	in-memory conversation store (create/add-message)	done	partial	unknown	unit	no
service	internal/roocode	task delegation (TaskSpec build; no real dispatch)	partial	no	unknown	unit	no
service	internal/roocode	code generation (Generate/Bootstrap scaffold)	partial	no	unknown	unit	no
service	internal/roocode	code review (file read + TODO/issue scan)	done	partial	unknown	unit	no
service	internal/roocode	i18n translator seam	done	yes	yes	unit	no
service	internal/telemetry	OpenTelemetry provider bootstrap (OTLP exporter)	done	yes	unknown	unit	no
service	internal/telemetry	agent instrumentation (spans + iteration counter)	done	yes	unknown	unit	no
service	internal/telemetry	LLM instrumentation (traced-provider wrapper)	done	yes	unknown	unit	no
service	internal/telemetry	tool instrumentation (call spans)	done	yes	unknown	unit	no
service	internal/telemetry	config-from-env (OTEL_* vars + exporter-kind resolve)	done	yes	yes	unit	no
infrastructure	internal/verifier	flexible timestamp parse (Unix + RFC3339)	done	yes	yes	unit	no
infrastructure	internal/verifier	adapter facade (cache+health+client, IsEnabled/Reachable)	done	yes	unknown	unit	no
infrastructure	internal/verifier	two-tier cache (TTL + Redis)	done	partial	unknown	unit	no
infrastructure	internal/verifier	health monitor (background liveness goroutine)	done	partial	unknown	unit	no
infrastructure	internal/verifier	background poller (interval model-sync)	done	partial	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
infrastructure	internal/verifier	bootstrap (embedded-or-remote stack init)	done	yes	unknown	unit,integ	no
infrastructure	internal/verifier	embedded verifier server (fallback)	partial	partial	unknown	unit	no
infrastructure	internal/verifier	fallback model list (offline mode)	done	yes	yes	unit	no
infrastructure	internal/verifier	real-server integration test (live LLMsVerifier)	done	yes	unknown	integ	no
service	internal/worker	consensus manager (Raft state machine, election/heartbeat)	done	partial	unknown	unit	no
service	internal/worker	vote-transport abstraction (decoupled from wire)	done	partial	unknown	unit	no
service	internal/worker	single-node fallback (nil-transport safe step-down)	done	yes	yes	unit	no
service	internal/worker	worker isolation: sandbox dir create (0750)	done	partial	unknown	unit	no
service	internal/worker	worker isolation: resource limits (cgroup writes)	done	partial	unknown	unit	no
service	internal/worker	SSH client pool (persistent connection reuse)	done	yes	yes	unit	no
service	internal/worker	consensus stress + chaos tests (concurrent votes, SIGKILL)	done	yes	yes	unit	no
service	internal/worker	worker-pool isolation stress test	done	yes	yes	unit	no
service	internal/server	Gin server + JWT auth middleware	done	yes	unknown	unit,integ	no
service	internal/server	project list/register handlers (user-from-context)	done	yes	yes	integ	no
service	internal/server	LLM generate endpoint (real provider stream)	done	yes	yes	unit,integ	no
service	internal/server	LLM model-list endpoint (verifier-queried)	done	yes	yes	unit,integ	no
service	internal/server	worker/task/project CRUD handlers	done	yes	yes	integ	no
service	internal/server	QA session start handler (helix_qa integration)	done	yes	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
service	internal/server	specify endpoint (spec from prompt)	done	yes	yes	unit,integ	no
service	internal/server	stats / pprof / error-categorize (400-vs-500) helpers	done	yes	yes	unit	no
service	internal/server	il8n translator seam (context-aware)	done	yes	yes	unit	no

Owned submodules (round-1 umbrella rows → deepened principal packages)

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
submodule	helix_agent	internal/llm multi-provider interface (Claude/DeepSeek/Gemini/Mistral/Qwen/xAI/OpenRouter/Ollama/llama.cpp/Bedrock/Azure)	done	yes	unknown	unit,integ	no
submodule	helix_agent	internal/ensemble multi-model debate/fusion orchestrator	done	yes	yes	unit,integ,e2e	no
submodule	helix_agent	internal/provider+providers per-provider abstraction layer	done	yes	unknown	unit	no
submodule	helix_agent	internal/verifier LLMsVerifier single-source-of-truth	done	yes	unknown	unit	no
submodule	helix_agent	internal/server HTTP(Gin) +WebSocket API (completions/ensemble/stream)	done	yes	unknown	unit,integ,e2e	no
submodule	helix_agent	internal/handlers per-route request/response handlers	done	yes	unknown	unit	no
submodule	helix_agent	internal/database PostgreSQL persistence (pgx/v5, migrations)	done	partial	unknown	integ	no
submodule	helix_agent		done	partial	unknown	integ	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
		internal/redis caching layer (go-redis/v9)					
submodule	helix_agent	internal/auth JWT + bcrypt/argon2 + OAuth2	done	partial	unknown	unit,integ	no
submodule	helix_agent	internal/security PII redaction + guardrails bridge	done	partial	unknown	unit	no
submodule	helix_agent	internal/streaming SSE/WebSocket token-level streaming	done	yes	unknown	unit	no
submodule	helix_agent	internal/cache semantic (embeddings dedup) caching	done	partial	unknown	unit	no
submodule	helix_agent	internal/knowledge RAG / vector-DB (zep) bridge	done	partial	unknown	unit	no
submodule	helix_agent	internal/monitoring Prometheus metrics	done	partial	unknown	unit	no
submodule	helix_agent	internal/logging structured JSON + gRPC tracing	done	partial	unknown	unit	no
submodule	helix_agent	internal/mcp Model Context Protocol server/client	done	partial	unknown	unit	no
submodule	helix_agent	internal/skills+plugins hot-reload plugin system	done	partial	unknown	unit	no
submodule	helix_agent	internal/analytics usage telemetry + warehouse export	partial	partial	unknown	unit	no
submodule	helix_agent	pkg/sdk Go + Python client SDKs	done	partial	unknown	unit	no
submodule	security	pkg/guardrails content guardrail engine (severity rules)	done	partial	unknown	unit	no
submodule	security	pkg/pii PII detect+redact	done	partial	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
		(email/phone/SSN/ CC-Luhn/IPv4)					
submodule	security	pkg/content composable filter chains (length/ pattern/keyword)	done	partial	unknown	unit	no
submodule	security	pkg/policy rule- based policy enforce (allow/ deny/audit)	done	partial	unknown	unit	no
submodule	security	pkg/scanner vuln- scan interface + report aggregation	done	partial	unknown	unit	no
submodule	security	pkg/headers HTTP security headers middleware (CSP/ HSTS/...)	done	partial	unknown	unit	no
submodule	security	pkg/securestorage AES-256-GCM + Argon2id key derivation	done	partial	unknown	unit	no
submodule	security	pkg/e2ee end-to- end encryption (ChaCha20- Poly1305)	done	partial	unknown	unit	no
submodule	security	pkg/ssrf SSRF prevention (outbound HTTP guard)	done	partial	unknown	unit	no
submodule	security	pkg/attestation + pkg/gpuattest platform/GPU attestation	partial	no	unknown	unit	no
submodule	helix_specifier	parallel execution (bounded- concurrency dispatch)	done	yes	unknown	unit	no
submodule	helix_specifier	constitution-as- code (machine- readable rule enforce)	done	yes	unknown	unit	no
submodule	helix_specifier	Nyquist TDD ($\geq 2x$ test-to-impl ratio enforce)	done	yes	unknown	unit	no
submodule	helix_specifier	debate architecture (multi-round spec refinement)	done	yes	yes	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
submodule	helix_specifier	skill learning (running-avg proficiency tracking)	done	yes	unknown	unit	no
submodule	helix_specifier	brownfield analysis (legacy pattern/dep detection)	done	yes	unknown	unit	no
submodule	helix_specifier	predictive specification (future-req prediction)	done	yes	unknown	unit	no
submodule	helix_specifier	cross-project transfer (shared knowledge base)	done	yes	unknown	unit	no
submodule	helix_specifier	adaptive ceremony (dynamic level by quality metrics)	done	yes	unknown	unit	no
submodule	helix_specifier	spec memory (persistent index + semantic search)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/autonomous autonomous QA session engine (~9.5k loc, 21 tests)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/orchestrator QA run orchestration	done	yes	unknown	unit	no
submodule	helix_qa	pkg/navigator UI navigation/driving (~3.8k loc, 16 tests)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/detector + pkg/issuedetector issue/error detection	done	yes	unknown	unit	no
submodule	helix_qa	pkg/recordingqa recording-validator (anti-bluff §11.4.107)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/regression standing regression-guard suite	done	yes	unknown	unit	no
submodule	helix_qa		done	yes	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
		pkg/evidence captured-evidence management					
submodule	helix_qa	pkg/challengegen Challenge generation	partial	yes	unknown	unit	no
submodule	helix_qa	pkg/reporter QA reporting	done	yes	unknown	unit	no
submodule	helix_qa	pkg/conduit + pkg/ bridge sync- channel / agent bridge (§11.4.116)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/replay + pkg/ reproduce replay / defect reproduction	done	partial	unknown	unit	no
submodule	helix_qa	pkg/distributed + pkg/maestro distributed/maestro coordination	partial	partial	unknown	unit	no
submodule	helix_qa	banks/ test-bank corpus (JSON/ YAML suites)	done	yes	unknown	none	no
submodule	helix_qa	cmd/* tooling (axtree/capture/ recvalidate/ omniparser/uitars/ lpips/...)	done	partial	unknown	unit	no
submodule	panoptic	internal/launcher multi-platform test launcher (web/ desktop/mobile)	done	no	unknown	unit	no
submodule	panoptic	internal/executor UI automation executor	done	no	unknown	unit	no
submodule	panoptic	internal/vision computer-vision element detection	done	no	unknown	unit	no
submodule	panoptic	internal/ocr OCR text extraction	done	no	unknown	unit	no
submodule	panoptic	internal/recvalidate recording validator	done	no	unknown	unit	no
submodule	panoptic	internal/ai AI test generation/ enhancement	done	no	unknown	unit	no
submodule	panoptic		done	no	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
submodule	panoptic	internal/cloud cloud storage/ integration internal/platforms per-platform automation adapters	done	no	unknown	unit	no
submodule	panoptic	screenshot + video recording capture	done	no	unknown	unit	no
submodule	panoptic	cmd/record + cmd/ vision + cmd/ testgen CLI tooling	done	no	unknown	unit	no

Round-2 coverage notes

- **Internal deepening:** the 12 round-1-flagged packages (clientcore, agentbridge, checkpoint, ensembleui, substrate, workspace, voice, roocode, telemetry, verifier, worker, server) are now broken out into 73 discrete feature rows. checkpoint, telemetry, most of verifier, server, and several worker features upgrade from round-1 partial to working-untaped on direct-source evidence (real git plumbing, real OTEL exporter, real Raft + cgroup writes, real Gin handlers with integration tests). roocode task-delegation/code-gen and voice/workspace auto-cleanup remain honest gap (real types, no shipped wiring / incomplete impl).
- **Stub-claim corrections** (see top caveat): three round-2-explorer "stub" reports were verified FALSE by direct read and corrected to done/partial. Logged here so the conductor does not propagate the wrong call.
- **Submodule Real-use=unknown throughout:** each submodule package compiles and has tests, but genuine end-user reachability THROUGH HelixCode (vs the submodule's own standalone tests) was not confirmed by static inspection — honestly unknown, never green.
- **helix_agent Wired:** yes for the packages HelixCode consumes via the direct dev.helix.agent import (llm/ensemble/server/provider/verifier/ streaming surface reachable through the agent dependency); partial for packages reachable only transitively (database/redis/auth/monitoring/etc).
- **panoptic Wired=no:** present as an equal-codebase submodule (CONST-051) but NOT in HelixCode's Go build graph — it is a Challenge/recording framework consumed out-of-process, not imported.
- Every Real-use=unknown / working-untaped row is a candidate for a recorded scenario; none is video-confirmed (📹 no throughout).

Round-2 deepened feature count: 145 rows (73 internal across 12 packages + 72 submodule across helix_agent 19, security 11, helix_specifier 10, helix_qa 14, panoptic 10, distributed across the wiring model).

Internal services + infrastructure

Inventory of every feature under `helix_code/internal/*` (72 packages). Assessed from code evidence (impl reality, wiring, `_test.go` presence) per CONST-035 / §11.4.107 anti-bluff. 📺 Video is no for every row (recordings are the conductor's job once a real analyzed recording exists); Overall is never confirmed for the same reason. Origin is native (HelixCode's own code).

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	📺 V
service	internal/adapters	il8n message resolution	done	yes	yes	unit	no	n
infrastructure	internal/adapters	translator injection seam	done	yes	yes	unit	no	n
infrastructure	internal/adapters	speckit debate adapter	partial	yes	unknown	unit	no	n
infrastructure	internal/adapters	container adapter	partial	yes	unknown	unit	no	n
service	internal/agent	agent execution interface	done	yes	yes	unit	no	n
service	internal/agent	agent capability management	done	yes	yes	unit	no	n
service	internal/agent	agent health check	done	yes	yes	unit	no	n
service	internal/agent	agent collaboration	done	yes	yes	unit	no	n
infrastructure	internal/agentbridge	verifier bridge	partial	yes	unknown	unit	no	n
service	internal/approval	approval request management	done	yes	yes	unit	no	n
service	internal/approval	approval mode selector	done	yes	yes	unit	no	n
infrastructure	internal/approvalwire	yes/no prompter	done	yes	yes	unit	no	n
service	internal/auth	user authentication	done	yes	yes	unit	no	n
service	internal/auth	session management	done	yes	yes	unit	no	n
service	internal/auth	password hashing (bcrypt/argon2)	done	yes	yes	unit	no	n
service	internal/auth	JWT token generation	done	yes	yes	unit	no	n
infrastructure	internal/autocommit	git auto-commit	done	yes	yes	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
infrastructure	internal/ autocommit	secret filter	done	yes	yes	unit	no	no
infrastructure	internal/cache	multi-tier cache	done	yes	yes	unit	no	no
infrastructure	internal/cache	redis tier	done	yes	yes	unit	no	no
infrastructure	internal/cache	disk tier	done	yes	yes	unit	no	no
infrastructure	internal/ checkpoint	checkpoint manager	partial	yes	unknown	unit	no	no
service	internal/ clarification	clarification engine	done	yes	yes	unit	no	no
service	internal/ clarification	question generation	done	yes	yes	unit	no	no
infrastructure	internal/ clientcore	agentic-tools provider	partial	yes	unknown	unit	no	no
infrastructure	internal/ clientcore	skills provider	partial	yes	unknown	unit	no	no
service	internal/ cognee	cognee client	done	yes	yes	unit	no	no
service	internal/ cognee	cognee manager	done	yes	yes	unit	no	no
infrastructure	internal/ cognee	cognee cache manager	done	yes	yes	unit	no	no
service	internal/ commands	aider command	done	yes	yes	unit	no	no
service	internal/ commands	approval command	done	yes	yes	unit	no	no
service	internal/ commands	browser command	done	yes	yes	unit	no	no
service	internal/ commands	edit command	done	yes	yes	unit	no	no
service	internal/ commands	git auto- commit command	done	yes	yes	unit	no	no
service	internal/ commands	mcp command	done	yes	yes	unit	no	no
service	internal/ commands	markdown commands	done	yes	yes	unit	no	no
service	internal/ commands	skills command	done	yes	yes	unit	no	no
service	internal/ commands	subagents command	done	yes	yes	unit	no	no
service		tasks command	done	yes	yes	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Verif
	internal/ commands							
service	internal/ commands	worktree command	done	yes	yes	unit	no	n
infrastructure	internal/ commands	command executor	done	yes	yes	unit	no	n
infrastructure	internal/ commands	command registry	done	yes	yes	unit	no	n
infrastructure	internal/ commands	command parser	done	yes	yes	unit	no	n
infrastructure	internal/config	viper-based config loader	done	yes	yes	unit	no	n
infrastructure	internal/config	cognee configuration management	done	yes	yes	unit	no	n
infrastructure	internal/config	verifier configuration	done	yes	yes	unit	no	n
infrastructure	internal/config	platform-UI adapters config	partial	yes	unknown	unit	no	n
infrastructure	internal/ context	context builder	done	yes	yes	unit	no	n
infrastructure	internal/ context	token counter	done	yes	yes	unit	no	n
infrastructure	internal/ context	history condenser	done	yes	yes	unit	no	n
service	internal/ continua	completion engine	done	yes	yes	unit	no	n
service	internal/ continua	continue-edit tool	done	yes	yes	unit	no	n
service	internal/ continua	continue- complete tool	done	yes	yes	unit	no	n
infrastructure	internal/ database	database connection pool	done	yes	yes	unit	no	n
infrastructure	internal/ database	postgres integration	done	yes	yes	integ	no	n
infrastructure	internal/ deployment	production deployer	done	yes	yes	unit	no	n
infrastructure	internal/ deployment	deployment strategy	done	yes	yes	unit	no	n
service	internal/ discovery	service registry with TTL + health checks	done	yes	yes	unit	no	n
service	internal/ discovery		done	yes	unknown	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	V
service	internal/discovery	UDP multicast broadcast discovery	done	yes	yes	unit	no	n
service	internal/discovery	dynamic port allocation	done	yes	yes	unit	no	n
service	internal/discovery	health monitoring (HTTP/gRPC/TCP)	done	yes	yes	unit	no	n
service	internal/editor	unified-diff code editing	done	yes	yes	unit	no	n
service	internal/editor	whole-file replacement editing	done	yes	yes	unit	no	n
service	internal/editor	search-and-replace editing	done	yes	yes	unit	no	n
service	internal/editor	line-range editing	done	yes	yes	unit	no	n
service	internal/editor	model-specific format selection	done	yes	yes	unit	no	n
service	internal/ensembleui	ensemble UI rendering	partial	partial	unknown	none	no	n
service	internal/event	publish-subscribe event bus	done	yes	yes	unit	no	n
service	internal/event	async + sync event handling	done	yes	yes	unit	no	n
service	internal/event	task/workflow/worker event types	done	yes	yes	unit	no	n
service	internal/fix	security issue auto-fix	done	yes	unknown	unit	no	n
service	internal/focus	hierarchical focus management	done	yes	unknown	unit	no	n
service	internal/focus	priority-based focus tracking	done	yes	unknown	unit	no	n
service	internal/focus	focus chain tracking	done	yes	unknown	unit	no	n
service	internal/hardware	CPU detection + profiling	done	yes	yes	unit	no	n
service	internal/hardware	GPU detection (NVIDIA/AMD/Apple/Intel)	done	yes	yes	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	V
service	internal/hardware	optimal LLM model-size inference	done	yes	yes	unit	no	n
service	internal/helixqa	HelixQA test wrapper	partial	yes	unknown	unit,integ	no	n
service	internal/hooks	hook registration + triggering	done	yes	yes	unit,integ	no	n
service	internal/hooks	priority-based hook execution	done	yes	yes	unit	no	n
service	internal/hooks	async/sync hook execution	done	yes	yes	unit	no	n
infrastructure	internal/i18nwiring	i18n catalog wire-all (multi-lang)	done	yes	yes	unit	no	n
infrastructure	internal/infraboot	on-demand infra boot (EnsureInfra)	done	yes	unknown	unit,integ	no	n
service	internal/kilocode	call-graph build + query	done	yes	unknown	unit	no	n
service	internal/kilocode	symbol rename engine	done	yes	unknown	unit	no	n
service	internal/kilocode	change-impact analyzer	done	yes	unknown	unit	no	n
service	internal/kilocode	refactor (extract-method/inline-call)	done	yes	unknown	unit	no	n
service	internal/kilocode	multi-edit tool	done	yes	unknown	unit	no	n
service	internal/llm	OpenAI provider (GPT models)	done	yes	yes	unit,integ	no	n
service	internal/llm	Anthropic provider (Claude models)	done	yes	yes	unit,integ	no	n
service	internal/llm	Google Gemini provider	done	yes	yes	unit	no	n
service	internal/llm	Azure OpenAI provider	done	yes	yes	unit	no	n
service	internal/llm	AWS Bedrock provider	done	yes	yes	unit	no	n
service	internal/llm	Ollama local provider	done	yes	yes	unit,integ	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
service	internal/llm	llama.cpp local inference	done	yes	yes	unit	no	no
service	internal/llm	Mistral provider	done	yes	yes	unit	no	no
service	internal/llm	DeepSeek provider	done	yes	yes	unit	no	no
service	internal/llm	Groq provider	done	yes	yes	unit	no	no
service	internal/llm	streaming response handling	done	yes	yes	unit,integ	no	no
service	internal/llm	token counting + accounting	done	yes	yes	unit	no	no
service	internal/llm	model discovery/ listing	done	yes	yes	unit	no	no
service	internal/llm	load balancing across providers	done	yes	yes	unit	no	no
service	internal/llm	provider health monitoring	done	yes	yes	unit	no	no
service	internal/llm	ensemble provider orchestration	done	yes	yes	unit	no	no
service	internal/llm	embeddings generation	done	yes	yes	unit	no	no
infrastructure	internal/logging	structured logging with levels	done	yes	yes	unit	no	no
infrastructure	internal/logging	named loggers	done	yes	yes	unit	no	no
infrastructure	internal/logo	logo processing + assets	done	yes	unknown	unit	no	no
infrastructure	internal/logo	color extraction + icon generation	done	yes	unknown	none	no	no
infrastructure	internal/mcp	Model Context Protocol server	done	yes	partial	unit,integ	no	no
infrastructure	internal/mcp	JSON-RPC 2.0 tool invocation	done	yes	yes	unit	no	no
infrastructure	internal/mcp	WebSocket session management	done	yes	yes	unit	no	no
infrastructure	internal/mcp		done	yes	unknown	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
		OAuth token management						
service	internal/memory	Cognee LLM memory integration	done	yes	unknown	unit,integ	no	no
infrastructure	internal/memory	memory state persistence	done	yes	unknown	unit	no	no
infrastructure	internal/monitoring	system metrics collection	done	yes	unknown	unit	no	no
infrastructure	internal/monitoring	health check monitoring	partial	partial	unknown	unit	no	no
service	internal/notification	multi-channel notification engine	done	yes	unknown	unit	no	no
service	internal/notification	Slack integration	done	yes	unknown	unit	no	no
service	internal/notification	Discord integration	done	yes	unknown	unit	no	no
service	internal/notification	Telegram integration	done	yes	unknown	unit	no	no
service	internal/notification	Email/SMTP integration	done	yes	unknown	unit	no	no
service	internal/notification	Teams integration	done	yes	unknown	unit	no	no
service	internal/notification	rate limiting + retry	done	yes	unknown	unit	no	no
infrastructure	internal/performance	performance optimizer	done	yes	unknown	unit	no	no
infrastructure	internal/performance	CPU optimization	partial	partial	unknown	unit	no	no
infrastructure	internal/performance	memory optimization	partial	partial	unknown	unit	no	no
infrastructure	internal/persistence	file-based state store	done	yes	unknown	unit	no	no
infrastructure	internal/persistence	session serialization	done	yes	unknown	unit	no	no
infrastructure	internal/persistence	auto-save manager	done	yes	unknown	unit	no	no
service	internal/planner	task executor	done	yes	unknown	unit	no	no
service	internal/planner	OpenHands plan execution	partial	partial	unknown	unit	no	no
infrastructure	internal/plantree	plan tree system	done	yes	unknown	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	V
infrastructure	internal/ plantree	plan node management	done	yes	unknown	unit	no	n
infrastructure	internal/ plantree	plan branching + merging	done	yes	unknown	unit	no	n
service	internal/ plugins	plugin base framework	done	yes	unknown	unit	no	n
service	internal/ plugins	plugin activation	done	yes	unknown	unit	no	n
service	internal/ plugins	plugin hooks registry	partial	partial	unknown	unit	no	n
service	internal/project	project manager	done	yes	unknown	unit	no	n
service	internal/project	project metadata	done	yes	unknown	unit	no	n
service	internal/project	database- backed storage	done	yes	unknown	unit	no	n
infrastructure	internal/ projectmemory	memory loader	done	yes	unknown	unit	no	n
infrastructure	internal/ projectmemory	filesystem watcher	done	yes	unknown	unit	no	n
infrastructure	internal/ projectmemory	memory registry	done	yes	unknown	unit	no	n
infrastructure	internal/ provider	provider interface	done	yes	unknown	unit	no	n
service	internal/ provider	multi-provider support	done	yes	unknown	unit	no	n
service	internal/ providers	AI integration	done	yes	unknown	unit,integ	no	n
service	internal/ providers	fallback + load balancing	done	yes	unknown	unit,integ	no	n
infrastructure	internal/ quality	quality gate scoring	done	yes	unknown	unit	no	n
infrastructure	internal/ quality	build verification	done	yes	unknown	unit	no	n
infrastructure	internal/ quality	linting validation	done	yes	unknown	unit	no	n
infrastructure	internal/redis	redis client	done	yes	unknown	unit	no	n
infrastructure	internal/redis	key-value operations	done	yes	unknown	unit	no	n
infrastructure	internal/redis	pub/sub messaging	done	yes	unknown	unit	no	n
service	internal/render	ANSI renderer	done	yes	unknown	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
service	internal/render	fancy-mode output	done	yes	unknown	unit	no	no
service	internal/render	frame buffer management	done	yes	unknown	unit	no	no
service	internal/render	streaming block renderer	done	yes	unknown	unit	no	no
infrastructure	internal/repomap	symbol extraction	done	yes	unknown	unit	no	no
infrastructure	internal/repomap	file ranking engine	done	yes	unknown	unit	no	no
infrastructure	internal/repomap	repo cache	done	yes	unknown	unit	no	no
infrastructure	internal/repomap	tree-sitter integration	done	yes	unknown	unit	no	no
service	internal/roocode	Roo-Code CLI port	partial	partial	unknown	unit	no	no
service	internal/roocode	template-based code generation	partial	partial	unknown	unit	no	no
service	internal/rules	rule hierarchy management	done	yes	yes	unit	no	no
service	internal/rules	rule pattern matching (glob/regex/exact)	done	yes	yes	unit	no	no
service	internal/rules	rule category + tag querying	done	yes	yes	unit	no	no
service	internal/secrets	secret loader from .env files	done	yes	yes	unit	no	no
service	internal/secrets	secret validation + missing-var detection	done	yes	yes	unit	no	no
service	internal/security	security manager (global + local)	done	yes	yes	unit	no	no
service	internal/security	feature scanning + zero-tolerance validation	done	yes	yes	unit	no	no
service	internal/server	auth endpoints (register/login/logout/refresh)	done	yes	unknown	unit	no	no
service	internal/server	user profile endpoints	done	yes	unknown	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
		(GET/PUT / me)						
service	internal/server	worker mgmt endpoints (list/register/heartbeat)	done	yes	unknown	unit	no	no
service	internal/server	task CRUD endpoints	done	yes	unknown	unit	no	no
service	internal/server	project CRUD endpoints	done	yes	unknown	unit	no	no
service	internal/server	workflow execution endpoints	done	yes	unknown	unit	no	no
service	internal/server	system stats + health-check endpoints	done	yes	unknown	unit	no	no
service	internal/server	LLM provider + model list endpoints	done	yes	unknown	unit	no	no
service	internal/server	memory system statistics endpoint	done	yes	unknown	unit	no	no
service	internal/server	WebSocket real-time communication	done	partial	unknown	unit	no	no
service	internal/session	session lifecycle (create/start/complete/pause)	done	yes	yes	unit	no	no
service	internal/session	session modes (planning/building/testing/refactoring)	done	yes	yes	unit	no	no
service	internal/session	focus chain integration	done	yes	yes	unit	no	no
service	internal/session	context + metadata tracking	done	yes	yes	unit	no	no
service	internal/session	session querying + filtering	done	yes	yes	unit	no	no
infrastructure	internal/substrate	substrate abstraction layer	partial	partial	unknown	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Verif
service	internal/task	task creation + assignment	done	yes	yes	unit	no	n
service	internal/task	task status lifecycle	done	yes	yes	unit	no	n
service	internal/task	priority-based task queue	done	yes	yes	unit	no	n
service	internal/task	checkpoint creation + recovery	done	yes	yes	unit	no	n
service	internal/task	dependency + circular-dependency validation	done	yes	yes	unit	no	n
service	internal/task	dependent task tracking	done	yes	yes	unit	no	n
service	internal/task	redis caching for tasks	done	yes	yes	unit	no	n
service	internal/telemetry	agent instrumentation (prompt/response)	done	partial	yes	unit	no	n
service	internal/telemetry	LLM instrumentation (calls/latency/tokens)	done	partial	yes	unit	no	n
service	internal/telemetry	tool instrumentation (execution/errors)	done	partial	yes	unit	no	n
service	internal/telemetry	OpenTelemetry provider integration	done	partial	yes	unit	no	n
service	internal/template	template creation + registration	done	yes	yes	unit	no	n
service	internal/template	variable substitution + validation	done	yes	yes	unit	no	n
service	internal/template	built-in template library	done	yes	yes	unit	no	n
service	internal/theme	theme detection (OS/system prefs)	done	yes	yes	unit	no	n
service	internal/theme		done	yes	yes	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Verif
		theme loading + customization						
service	internal/theme	built-in theme collection	done	yes	yes	unit	no	n
service	internal/tools	file read/write operations	done	yes	yes	unit	no	n
service	internal/tools	file editing with string replacement	done	yes	yes	unit	no	n
service	internal/tools	file globbing (pattern matching)	done	yes	yes	unit	no	n
service	internal/tools	file content search (grep)	done	yes	yes	unit	no	n
service	internal/tools	shell command execution	done	yes	yes	unit	no	n
service	internal/tools	background shell execution	done	yes	yes	unit	no	n
service	internal/tools	web page fetching + parsing	done	yes	unknown	unit	no	n
service	internal/tools	web search integration	done	yes	unknown	unit	no	n
service	internal/tools	browser automation (launch/navigate/screenshot)	done	yes	unknown	unit	no	n
service	internal/tools	codebase mapping + symbol definitions	done	yes	yes	unit	no	n
service	internal/tools	multi-file transactional editing	done	yes	yes	unit	no	n
service	internal/tools	user confirmation prompts	done	yes	yes	unit	no	n
service	internal/tools	notebook read/edit operations	done	yes	yes	unit	no	n
infrastructure	internal/tools	tool registry + execution	done	yes	yes	unit	no	n
service	internal/verifier	LLMsVerifier HTTP client	done	yes	unknown	unit,integ	no	n
service			done	yes	unknown	unit	no	n

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Timeline
		model metadata adapter						
infrastructure	internal/verifier	two-tier cache (LRU + Redis)	done	partial	unknown	unit	no	no
infrastructure	internal/verifier	health monitoring + circuit breaker	done	partial	unknown	unit	no	no
infrastructure	internal/verifier	background poller (real-time updates)	done	partial	unknown	unit	no	no
infrastructure	internal/verifier	event publishing to HelixCode bus	done	partial	unknown	unit	no	no
service	internal/version	version string retrieval	done	yes	yes	unit	no	no
service	internal/version	build metadata exposure	done	yes	yes	unit	no	no
service	internal/voice	audio capture (arecord/sox)	partial	partial	unknown	unit	no	no
service	internal/voice	Whisper API transcription	partial	partial	unknown	unit	no	no
service	internal/voice	local whisper.cpp fallback	partial	no	unknown	unit	no	no
service	internal/worker	worker registration + management	done	yes	yes	unit	no	no
service	internal/worker	SSH-based remote execution	done	yes	yes	unit	no	no
service	internal/worker	health monitoring + metrics	done	yes	yes	unit	no	no
service	internal/worker	capability auto-detection	done	yes	yes	unit	no	no
service	internal/worker	worker isolation + sandboxing	done	partial	unknown	unit	no	no
service	internal/worker	consensus protocol (leader election)	done	partial	unknown	unit	no	no
service	internal/worker	host-key verification + SSH security	done	yes	yes	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	Review
service	internal/workflow	planning workflow execution	done	yes	unknown	unit	no	no
service	internal/workflow	building workflow execution	done	yes	unknown	unit	no	no
service	internal/workflow	testing workflow execution	done	yes	unknown	unit	no	no
service	internal/workflow	refactoring workflow execution	done	yes	unknown	unit	no	no
service	internal/workflow	DAG-based step orchestration	done	yes	yes	unit	no	no
service	internal/workflow	LLM provider integration for workflows	done	partial	unknown	unit	no	no
infrastructure	internal/workspace	container-based workspace management	partial	partial	unknown	unit	no	no
infrastructure	internal/workspace	Docker/Podman container orchestration	partial	partial	unknown	unit	no	no
infrastructure	internal/workspace	project directory mounting + isolation	partial	partial	unknown	unit	no	no
infrastructure	internal/workspace	auto-cleanup TTL enforcement	partial	no	unknown	unit	no	no
infrastructure	internal/mocks	unit-test mock fixtures (test-only, not a user feature)	done	yes	n/a	none	no	no
infrastructure	internal/pproftutil	pprof profiling helper (dev/test support)	done	partial	unknown	none	no	no
infrastructure	internal/testutil	shared test utilities (test-only, not a user feature)	done	yes	n/a	none	no	no

Coverage notes

- **Partially assessed (need deeper inventory):** clientcore, agentbridge, checkpoint, ensembleui, substrate, workspace, voice, roocode, telemetry, verifier (poller/event-bus wiring), worker (isolation/consensus), and server real-use (declared endpoints, no integration evidence inspected). Their core types compile and have unit tests, but wiring into a shipped flow and genuine end-user reachability could not be fully confirmed from static inspection alone — flagged partial/unknown honestly rather than green.
- mocks, pprofutil, testutil are dev/test-support packages, not user features — listed for completeness, marked n/a.
- Every Real-use=unknown and every working-untaped row is a candidate for a recorded scenario; none is video-confirmed yet (📺 no throughout).

233 features inventoried across 72 packages.

Ported cli_agents capabilities

Evidence-backed inventory of capabilities **actually ported into HelixCode** from the cli_agents/ reference catalogue (51 vendored reference agents). Per CONST-035 anti-bluff: this lists **ONLY** capabilities with landed code evidence (package doc.go origin headers, CONTINUATION.md P2-Fxx CLOSED ledger, POWER_FEATURES_PORTING_PLAN rev2 file:line reconciliation, git port commits) — NOT every cli_agent's full feature set. Planned-but-not-landed items are marked Dev=absent/partial honestly. **No feature is 📺 yes** (no recordings analyzed for ported features). **Overall is never confirmed** (a confirmed rollup requires an analyzed video).

Evidence basis: (1) CONTINUATION.md ports ledger lists P2-F21..P2-F30 as CLOSED with landing packages; (2) each landed package's doc.go names its source agent ("aider voice input port P2-F27", "Roo-code CLI agent port", "Continue.dev IDE integration", "Go port of upstream gptme's profile mechanism"); (3) POWER_FEATURES_PORTING_PLAN.md rev2 reconciles Phase-1 against HEAD with file:line; (4) HXC-031-codex-cline-port.md is DRAFT (plan only, NO code) — codex multimodal + cline computer-use are PLANNED, not landed.

Area	Component (HelixCode pkg)	Feature	Dev	Wired	Real-us
service	internal/approval	Approval/exec-policy modes (per-tool yes/no, autopilot tiers)	done	yes	unknow
service	internal/autocommit	Git auto-commit per change (generated msg, secret-filter, summariser)	done	yes	unknow
service	internal/tools/browser	Browser tool suite (launch/click/type/scroll/screenshot via chromedp)	done	yes	unknow
service	internal/projectmemory	Project-memory context files (AGENTS.md/.clinerules-style loader+watcher)	done	yes	unknow
service	internal/plantree		done	yes	unknow

Area	Component (HelixCode pkg)	Feature	Dev	Wired	Real-us
		Plan trees (branching persistent implementation plans, JSON-backed)			
service	internal/session (condense.go)	Context compaction / history condense (CompactIfNeeded/ShouldCompact)	done	yes	unknown
service	internal/workspace	Container per-task workspace (Docker/Podman mount, TTL cleanup)	done	partial	unknown
service	internal/voice	Voice-to-code input (arecord/sox capture, Whisper API + whisper.cpp fallback)	done	yes	unknown
service	internal/repomap	Repo-map (tree-sitter ranked incremental project map)	done	yes	unknown
service	internal/kilocode	AST-aware refactoring (cross-file rename, impact/call-graph, extract/move/inline)	done	yes	unknown
service	internal/roocode	Roo-code full port (task delegation, template gen, diff review, conv memory)	done	yes	unknown
application	internal/continua	Continue.dev IDE integration (inline completions, editor, chat panel, diff, model selector)	done	partial	unknown
service	internal/agent/profiles	Verifier work profile + RoleVerify posture (subagent review/validation)	done	yes	unknown
application	cmd/cli (main.go:2118,2326) + autocommit/git.go	/undo + /diff (git-aware, force-push-free revert)	done	yes	unknown
service	internal/workflow/ autonomy	Configurable autonomy presets (None/Basic/BasicPlus/SemiAuto/FullAuto, 5 tiers)	done	yes	unknown
service	internal/workflow/ planmode	First-class Plan/Act mode controller (tool-gated by mode)	done	yes	unknown
service	internal/tools/askuser	ask_user interactive clarification tool	done	yes	unknown
service	internal/commands (markdown_skills.go)	Markdown skills subsystem (project>user 2-tier precedence)	partial	yes	unknown
service	internal/checkpoint	Workspace file-snapshot checkpoint + /checkpoint create/list/restore	done	yes	unknown
application	cmd/cli (main.go:1727)	Per-request token-usage counter (real provider Usage, anti-fabrication)	done	yes	unknown
service	(no production code)	Context-window-% indicator	absent	no	no
service	(no production code)		partial	partial	no

Area	Component (HelixCode pkg)	Feature	Dev	Wired	Real-us
		TODO/step tracker surface (/tasks is background-job list, not TODO tracker)			
service	markdown_skills.go	SKILL.md built-in/bundled tier + canonical SKILL.md filename	absent	no	no
service	internal/llm (HXC-031 plan)	Codex multimodal image-content LLM request surface	absent	no	no
service	internal/tools/browser (HXC-031 plan)	Cline computer-use feedback loop (screenshot-per-action coord control)	absent	no	no
service	(planned, POWER_FEATURES F22)	Cumulative diff-review sandbox (stage-before-apply, apply/reject hunks)	absent	no	no
service	(planned, POWER_FEATURES F25)	Conversation branches / fork	absent	no	no
service	(planned, POWER_FEATURES F36/F43)	/rewind + Tangent mode	absent	no	no
service	(planned, POWER_FEATURES F18)	Messaging connectors (Slack/Telegram/Discord)	absent	no	no
service	(planned, POWER_FEATURES F35)	ACP mode (Agent Client Protocol over stdio)	absent	no	no
service	(planned, POWER_FEATURES F56)	OpenAI-compatible server endpoints	absent	no	no
service	(planned, POWER_FEATURES F61)	Spec-driven workflow surface (specify→plan→tasks→implement)	absent	no	no
service	(planned, POWER_FEATURES F33)	OS-level exec sandbox (Seatbelt/Landlock/bwrap)	absent	no	no

Count: 33 rows total — **20 landed ports** (Dev=done, working-untaped), **3 partial-landed** (workspace, continua, markdown-skills), **10 planned/not-landed** (Dev=absent/partial, gap). Zero 🚧 yes; zero confirmed.

Honest assessment: The **Phase-2 port wave (P2-F21..F30) genuinely landed** — 10 named source-agent ports (codex/aider/cline/plandex/openhands/kilo-code/roo-code/continua) ship as real packages with origin-attributed doc.go headers, unit tests, and CONTINUATION CLOSED records, plus the gptme verifier-profile port. The **POWER_FEATURES Phase-1 wrap (autonomy/Plan-Act/undo-diff/ask_user/skills/ token-count/checkpoint) is largely landed** per the rev2 file:line reconciliation. Beyond that, the porting is **mostly PLANNED, not landed**: POWER_FEATURES_PORTING_PLAN is explicitly "DRAFT — research + plan

only, NO code", its Phases 2–7 (diff-sandbox, branches, rewind, tangent, connectors, ACP, OpenAI server, spec-driven, OS sandbox) are unimplemented, and HXC-031 (codex multimodal + cline computer-use) is a DRAFT plan with no code. No ported feature has an analyzed recording, so none can be confirmed; all landed ports are honestly working–untaped pending video V&V.

Owned-submodule capabilities

Inventory of the discrete features each owned (own-org vasic-digital / HelixDevelopment) submodule under submodules/* provides to HelixCode. Assessed from each submodule's README.md, exported package surface, and *_test.go presence (CONST-035 / §11.4.107 — evidence-based, no bluffs).

Wiring model (load-bearing). HelixCode's inner Go module (helix_code/go.mod) replaces + imports only 3 submodules directly — helix_agent (dev.helix.agent), dag_orchestrator (dev.helix.dag), pipeline_runtime (dev.helix.pipeline). HelixCode source additionally imports a handful by module path (containers, helixspecifier, helixqa, concurrency, helixmemory, debate, lazy, llmsverifier, memory, llmprovider). A large set is wired **transitively** because helix_agent/go.mod requires ~35 own-org submodules — these are reachable through the agent dependency but NOT imported by HelixCode directly (Wired=partial). The remainder are present in the repo as equal-codebase submodules (CONST-051) but are NOT in HelixCode's Go build graph at all (Wired=no). 📺 Video=no for every row (conductor owns video confirmation; nothing here is confirmed). Origin=native throughout (all own-org).

Area	Component	Feature	Dev	Wired	Real-use	Te
submodule	helix_agent	Ensemble multi-LLM agent service (response fusion)	done	yes	yes	un
submodule	helix_agent	ReAct/tool-calling agent runtime (consumed by HelixCode internal/agent)	done	yes	yes	un
submodule	helix_agent	Skill registry + debate + verifier wiring (umbrella, 1583 test files)	done	yes	unknown	un
submodule	dag_orchestrator	Agent-free DAG scheduler (topo dispatch, worker pool, retry/backoff)	done	yes	yes	un
submodule	pipeline_runtime	Staged streaming dataflow runtime (push operators + FBP backpressure)	done	yes	yes	un
submodule	helix_specifier	Spec-driven dev fusion (SpecKit 7-phase + TDD + GSD milestones)	done	partial	unknown	un
submodule	helix_specifier	10 power features (parallel exec, constitution-as-code, debate phases)	partial	partial	unknown	un

Area	Component	Feature	Dev	Wired	Real-use	Te
submodule	helix_memory	Unified cognitive memory (Mem0+Cognee+Letta+Graphiti fusion)	done	yes	unknown	un
submodule	helix_memory	Parallel cross-backend search + 3-stage fusion/dedup/re-rank	done	yes	unknown	un
submodule	debate_orchestrator	Multi-agent debate orchestration (consensus + dissent, LessonBank)	done	yes	unknown	un
submodule	debate_orchestrator	Aux packages (validation/audit/evaluation/reflexion/tools)	stub	partial	no	un
submodule	concurrency	Worker pools, priority queues, rate limiters, circuit breakers, semaphores	done	yes	yes	un
submodule	lazy	Type-safe lazy-init primitives (sync.Once generics)	done	yes	yes	un
submodule	memory	Mem0-style scoped memory + entity extraction + knowledge graph + leak detect	done	partial	unknown	un
submodule	llm_provider	LLMProvider interface + circuit breaker + health monitor + retry + lazy	done	partial	unknown	un
submodule	llms_verifier	LLM verification platform (existence/latency/streaming/vision/embeddings)	done	yes	unknown	un
submodule	llms_verifier	Single-source-of-truth model/provider metadata (CONST-036/037)	done	yes	unknown	un
submodule	containers	Container lifecycle (boot/compose/health) for infra-on-demand (§11.4.76)	done	yes	unknown	un
submodule	helix_qa	Autonomous QA / Challenge orchestration (test-bank runner) [QA infra]	done	yes	unknown	un
submodule	panoptic	Recording-validator / observation harness [QA infra]	done	partial	unknown	un
submodule	agentic	Graph-based agentic workflow engine (branch/checkpoint/self-correct)	done	partial	unknown	un
submodule	auth	JWT / API-key / OAuth2 / HTTP auth middleware / token store	done	partial	unknown	un
submodule	cache	Multi-backend cache (mem/Redis/PG) + distributed patterns + TTL/eviction	done	partial	unknown	un
submodule	storage	Object storage (S3/MinIO/local) + cloud credential mgmt	done	partial	unknown	un

Area	Component	Feature	Dev	Wired	Real-use	Test
submodule	vector_db	Unified vector store (Qdrant/ Pinecone/Milvus/pgvector)	done	partial	unknown	unknown
submodule	embeddings	Text embeddings across 7 providers (OpenAI/Cohere/ Voyage/Jina/Google/Bedrock)	done	partial	unknown	unknown
submodule	event_bus	Typed pub/sub bus + glob/ prefix/metadata filtering + middleware	done	partial	unknown	unknown
submodule	messaging	Message-broker abstraction (in- mem/Kafka/RabbitMQ) + producer/consumer patterns	done	partial	unknown	unknown
submodule	streaming	SSE / WebSocket(rooms) / gRPC streaming / webhook(HMAC) / HTTP+breaker / Gin	done	partial	unknown	unknown
submodule	rag	RAG pipeline (3 chunkers, BM25+semantic+hybrid retrieve, MMR rerank, RRF fusion)	done	partial	unknown	unknown
submodule	tool_schema	Tool schema/validation/exec + 14 built-in safe tool handlers	done	partial	unknown	unknown
submodule	plugins	Plugin lifecycle (dep-ordered) + .so/process loaders + sandbox + output parse	done	partial	unknown	unknown
submodule	planning	AI planning algorithms (HiPlan + MCTS + Tree-of-Thoughts)	done	partial	unknown	unknown
submodule	optimization	Semantic cache + prompt compression + structured-output + SGLang/LlamaIndex/ LangChain	done	partial	unknown	unknown
submodule	self_improve	RLAIF pipeline (reward models, feedback, policy/ prompt optimization)	done	partial	unknown	unknown
submodule	benchmark	LLM benchmarking (SWE- bench/HumanEval/MMLU/ GSM8K) + leaderboard	done	partial	unknown	unknown
submodule	llm_ops	LLMOps (continuous eval, A/B experiments, prompt versioning, alerting)	done	partial	unknown	unknown
submodule	mcp_module	MCP server+client (JSON-RPC 2.0 stdio+HTTP/SSE) + adapter registry	done	partial	unknown	unknown
submodule	models	Shared AI/LLM data types (LLMRequest/Response, MCP/ LSP/ACP, code-intelligence)	done	partial	unknown	unknown
submodule	formatters		done	partial	unknown	unknown

Area	Component	Feature	Dev	Wired	Real-use	Te
		Pluggable code-formatter registry + engine + cache + native-binary shims				
submodule	vision_engine	CV + LLM-vision UI analysis + navigation-graph construction	partial	partial	unknown	un
submodule	doc_processor	Doc-feature-map extraction + verification-coverage tracking (QA-oriented)	done	partial	unknown	un
submodule	security	Security tooling module (round-300 deep-doc) — needs deeper inventory	done	partial	unknown	un
submodule	red_team	YAML-driven adversarial-prompt fixture harness (defensive guardrail regression)	done	partial	unknown	un
submodule	llm_orchestrator	Headless CLI-agent orchestration (OpenCode/ Claude/Gemini/Junie/Qwen)	done	no	no	un
submodule	helix_llm	Distributed LLM service (OpenAI/Anthropic APIs, llama.cpp, RAG, ReAct, HTTP/ 3)	done	no	no	un
submodule	conversation	Infinite-context (event-sourcing) + LLM compression + LRU cache	done	no	no	un
submodule	database	Driver-agnostic DB (PG/ SQLite) + pool + migrations + repo + query builder	done	no	no	un
submodule	document	Document model (18-format detect, change tracking, JSON serialize)	done	no	no	un
submodule	filesystem	Multi-protocol FS client (SMB/ FTP/NFS/WebDAV/Local)	done	no	no	un
submodule	config	Config mgmt (JSON files, env binding, validation, 8 storage-protocol types)	done	no	no	un
submodule	i18n	i18n library (bundles, loader, HTTP middleware)	done	no	no	un
submodule	observability	Distributed tracing + metrics + structured logging + health + analytics	done	no	no	un
submodule	middleware	Reusable net/http middleware (requestid/logging/recovery/ cors/chain)	done	no	no	un
submodule	rate_limiter	Rate limiting (in-mem + Redis) + HTTP middleware	done	no	no	un
submodule	recovery		done	no	no	un

Area	Component	Feature	Dev	Wired	Real-use	Te
		Named circuit breakers + periodic health checks + resilience facade				
submodule	watcher	Filesystem change monitoring (debounce, filters, handler chains)	done	no	no	un
submodule	background_tasks	Persistent task queue (PG) + worker pool + stuck-detect + DLQ + progress	done	no	no	un
submodule	skill_registry	Skill mgmt (load YAML/JSON/MD, register, execute, validate, store)	done	no	no	un
submodule	embeddings	(see embeddings row — module also standalone)	—	—	—	—
submodule	auto_temp	LLM temperature auto-tuning (multi-temp run + multi-judge scoring)	done	no	no	un
submodule	hyper_tune	LLM hyperparameter optimization (random/grid/Bayesian-lite)	done	no	no	un
submodule	i_llm	Structured-reasoning patterns (CoT/ToT/ReAct/few-shot/prompt-chain)	done	no	no	un
submodule	veritas	AI-truthfulness verification, fact-check, hallucination detection	done	no	no	un
submodule	leak_hub	System-prompt-leak detection + searchable archive	done	no	no	un
submodule	claritas	Defensive guardrail (leaked-prompt archive + extraction-attempt detector)	done	no	no	un
submodule	gandalf_solutions	Read-only Gandalf prompt-hacking solutions archive (research)	done	no	no	un
submodule	ouroborous	Self-referential AI-safety (recursive self-improve, runaway-loop detect)	done	no	no	un
submodule	normalize	Adversarial-input canonicalization (base64/leet/homoglyph/NFKC/ROT13...)	done	no	no	un
submodule	plinius_common	Plinius shared lib (config validators, error types, gRPC client, i18n, types)	done	no	no	un
submodule	toon	Token-Oriented Object Notation encode/decode	stub	no	no	un

55 features across 50 submodules.

Coverage-depth honesty

- **Shallow-inventoried (principal features only; deeper inventory warranted):** `helix_agent` (1583 test files — only top umbrella capabilities captured; internal package-level features not enumerated), `helix_qa` (646 test files, QA infra — single capability row), `helix_specifier` (37 tests, 10 power features collapsed to 2 rows), `security` (deep-doc module — only a single umbrella row; per-package security capabilities NOT enumerated, flagged as needs-deeper-inventory), `panoptic` (QA infra — single row).
- **Real-use is unknown** for nearly every transitively-wired (partial) and standalone (no) submodule because reachability through `helix_agent`'s `go.mod` $\neq$ proof an end user exercises the capability via `HelixCode`; only the 3 directly-replaced modules + `concurrency/lazy` show yes.
- **Dev reflects README claims + test presence**, NOT runtime verification. `toon` is honestly stub (PENDING_IMPLEMENTATION per its own README); `debate_orchestrator` aux packages are stub (NotYetImplemented per ACK-STUB).
- **Tests column** marks unit baseline (all 70 own-org submodules have `*_test.go`); `integ/e2e` added only where README/dir evidence shows real infra/integration suites (`cache`, `storage`, `vector_db`, `llms_verifier`, `helix_qa`, `helix_agent`, `containers`, `database`, `rate_limiter`, `helix_llm`, `observability-N/A`).
- Infra/tooling submodules excluded from feature rows per task scope: `docs_chain`, `challenges` (and `containers/helix_qa/panoptic` included only as the capabilities `HelixCode` consumes, marked QA/infra).

Inventory sources

- `docs/features/inventory/cmd_and_clients.md`
- `docs/features/inventory/deepened_round2.md`
- `docs/features/inventory/internal_services.md`
- `docs/features/inventory/ported_cli_agents.md`
- `docs/features/inventory/submodules.md`